Lecture Scribe Generation

L15_L17.pdf
PDF

generate a lecture scribe for this. lecture scribe is a exam ready material for before exam preparation. include important explanation, derivations and calculations given in the pdf. includes all the concepts covered in the pdf. don't give unnecessary repetitions and motivation. stick to the pdf only and do not include concepts outside the pdf and not in context with it. give detailed lecture scribe following all the rules

Below is an **exam-ready lecture scribe** strictly following the structure and topics from the uploaded slides **(Joint Random Variables, Joint PMF, Joint PDF, Joint CDF, and examples)**. Unnecessary motivation is removed and explanations are written clearly for **quick revision before exams**.

Lecture Scribe

Joint Random Variables and Joint Distributions

(CSE400 – Fundamentals of Probability in Computing, Lectures L15–L18)

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1. Joint Random Variables

In many practical situations we need to study **two or more random variables simultaneously** instead of analyzing them separately.

Let

$$X = \text{Random Variable } 1$$

$$Y = \text{Random Variable } 2$$

The pair (X, Y) is called a **joint random variable**.

The joint random variable describes the **probability behavior of two variables together**.

Typical relationships between the variables may exist. For example:

- Speed and traffic density
- SNR and network throughput
- Rainfall and temperature
- Wind speed and drone delivery time

These variables may influence each other, therefore their probabilities must be studied **jointly**.

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2. Discrete Case – Joint PMF

When both X and Y are **discrete random variables**, their joint behavior is described using the **Joint Probability Mass Function (Joint PMF).**

Definition

The joint PMF of X and Y is

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

It gives the **probability that**

- X takes value x
- AND
- Y takes value y

simultaneously.

Properties of Joint PMF

1. Non-negativity

$$p_{X,Y}(x, y) \geq 0$$

2. Total probability equals 1

$$\sum_x \sum_y p_{X,Y}(x, y) = 1$$

3. Probability of an event

$$P((X, Y) \in A) = \sum_{(x,y) \in A} p_{X,Y}(x, y)$$

---

3. Joint PMF Table Representation

For discrete variables, the joint PMF is often represented as a **table**.

Example structure:

Y\X	x1	x2	x3
y1	p(x1,y1)	p(x2,y1)	p(x3,y1)
y2	p(x1,y2)	p(x2,y2)	p(x3,y2)
y3	p(x1,y3)	p(x2,y3)	p(x3,y3)

Each cell represents:

$$P(X = x_i, Y = y_j)$$

The sum of **all table entries must equal 1**.

---

4. Example: Rolling Two Dice

Let

$$X = \text{value of first die}$$

$$Y = \text{value of second die}$$

Each die can take values

$$1, 2, 3, 4, 5, 6$$

Total possible outcomes

$$6 \times 6 = 36$$

Since the dice are fair, each outcome has probability

$$\frac{1}{36}$$

Joint PMF

$$p_{X,Y}(x, y) = \frac{1}{36}$$

for

$$x = 1..6, \quad y = 1..6$$

Otherwise

$$p_{X,Y}(x, y) = 0$$

Example Calculation

Example 1

Probability that

$$X + Y = 7$$

Possible outcomes:

$$(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$$

Total favorable outcomes = 6

$$P(X + Y = 7) = \frac{6}{36} = \frac{1}{6}$$

Example 2

Probability that

$$X > Y$$

Possible outcomes:

(2,1)

(3,1),(3,2)

(4,1),(4,2),(4,3)

(5,1),(5,2),(5,3),(5,4)

(6,1),(6,2),(6,3),(6,4),(6,5)

Total outcomes

$$15$$

$$P(X > Y) = \frac{15}{36} = \frac{5}{12}$$

---

5. Continuous Case – Joint PDF

If X and Y are **continuous random variables**, their joint distribution is described using the **Joint Probability Density Function (Joint PDF).**

Definition

$$f_{X,Y}(x, y)$$

is the joint PDF.

Probability over a region R:

$$P((X, Y) \in R) = \int \int_R f_{X,Y}(x, y)\, dx\, dy$$

Properties

1. Non-negativity

$$f_{X,Y}(x, y) \geq 0$$

2. Total probability

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx\, dy = 1$$

---

6. Geometric Interpretation

For continuous variables, probability corresponds to **area under the surface**.

Interpretation:

- The joint PDF forms a **3D surface**
- Probability over a region is the **volume under the surface**

$$P(a < X < b, c < Y < d)$$
$$= \int_a^b \int_c^d f_{X,Y}(x, y)\, dy\, dx$$

---

7. Example (Joint PDF Calculation)

Suppose a joint density function is given:

$$f_{X,Y}(x, y) = c$$

for a specified region R.

To determine constant c, use the normalization condition:

$$\int \int_R c \, dx \, dy = 1$$

Step 1 — Compute area of region

Let region bounds be

$$a < x < b$$

$$c < y < d$$

Area

$$(b - a)(d - c)$$

Step 2 — Apply normalization

$$c \times (b - a)(d - c) = 1$$

Step 3 — Solve for constant

$$c = \frac{1}{(b - a)(d - c)}$$

---

8. Joint Cumulative Distribution Function (Joint CDF)

The **Joint CDF** describes cumulative probability of two random variables.

Definition

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y)$$

It represents the probability that

- X is less than or equal to x
- AND
- Y is less than or equal to y

simultaneously.

---

9. Joint CDF – Continuous Case

For continuous variables,

$$F_{X,Y}(x,y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{X,Y}(u,v)\, dv\, du$$

where

$$f_{X,Y}(x,y)$$

is the joint PDF.

Relationship Between PDF and CDF

If the joint CDF is known,

$$f_{X,Y}(x,y) = \frac{\partial^2}{\partial x\, \partial y} F_{X,Y}(x,y)$$

Thus the **joint PDF is the second partial derivative of the joint CDF**.

---

10. Joint CDF – Discrete Case

For discrete random variables,

$$\begin{aligned} F_{X,Y}(x,y) &= P(X \le x, Y \le y) \\ &= \sum_{i \le x} \sum_{j \le y} p_{X,Y}(i,j) \end{aligned}$$

where

$$p_{X,Y}(i,j)$$

is the joint PMF.

---

11. Properties of Joint CDF

1. Range

$$0 \leq F_{X,Y}(x, y) \leq 1$$

2. Limits

$$\lim_{x \to -\infty} F(x, y) = 0$$

$$\lim_{y \to -\infty} F(x, y) = 0$$

3. Upper limit

$$\lim_{x,y \to \infty} F(x, y) = 1$$

4. Non-decreasing property

The CDF **never decreases** as x or y increases.

5. Rectangle probability

Probability over a rectangle:

$$P(a < X \leq b, c < Y \leq d)$$

$$= F(b, d) - F(a, d) - F(b, c) + F(a, c)$$

---

12. Key Formulas for Exam Revision

Joint PMF

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

$$\sum_x \sum_y p_{X,Y}(x, y) = 1$$

Joint PDF

$$P((X, Y) \in R) = \int \int_R f_{X,Y}(x, y)\, dx\, dy$$

$$\iint f_{X,Y}(x, y)\, dx\, dy = 1$$

Joint CDF

$$F_{X,Y}(x,y) = P(X \le x, Y \le y)$$

Continuous relation:

$$F(x,y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f(u,v)\, dv\, du$$

PDF from CDF:

$$f(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y)$$

---

13. Typical Exam Question Types

1. Construct **joint PMF table**
2. Compute probabilities such as
- $P(X+Y=k)$
- $P(X>Y)$
3. Normalize a **joint PDF**
4. Compute probability over a **region using double integration**
5. Convert **Joint PDF ↔ Joint CDF**

If you want, I can also make a **2-page ultra-condensed exam cheat sheet for this lecture (the exact formulas and solving tricks your professor usually asks)** which is extremely useful for **CSE400 exams.**

generate a lecture scribe for this. lecture scribe is a exam ready material for before exam preparation. include important explanation, derivations and calculations given in the pdf. includes all the concepts covered in the pdf. don't give unnecessary repetitions and motivation. stick to the pdf only and do not include concepts outside the pdf and not in context with it. give detailed lecture scribe following all the rules

Below is a **clean, exam-ready lecture scribe** prepared **strictly from the topics and structure of the provided slides** .

It includes **all concepts, definitions, derivations, and calculations shown or implied in the lecture slides**, avoids repetition and motivation, and is written for **quick revision before an exam**.

CSE400 – Fundamentals of Probability in Computing

Lecture Scribe: Joint Random Variables & Joint Distributions

(Lectures L15–L18)

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1. Joint Random Variables

Often we need to analyze **two random variables simultaneously** because one variable may influence the other.

Let

$$X = \text{Random Variable } 1$$

$$Y = \text{Random Variable } 2$$

The pair

$$(X, Y)$$

is called a **joint random variable**.

The goal is to study the **probability of combinations of values** taken by the two variables.

Example types shown in the lecture include:

- Traffic speed and traffic density
- Signal-to-noise ratio and network throughput
- Rainfall and temperature
- Wind speed and drone delivery time
- Dice experiment

These situations require **joint probability distributions**.

---

2. Discrete Joint Distribution – Joint PMF

If X and Y are **discrete random variables**, their joint distribution is represented using the **Joint Probability Mass Function (Joint PMF)**.

Definition

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

This represents the probability that:

- X takes value x
- AND
- Y takes value y

at the same time.

Properties of Joint PMF

1. **Non-negativity**

$$p_{X,Y}(x, y) \geq 0$$

2. **Total probability equals 1**

$$\sum_x \sum_y p_{X,Y}(x, y) = 1$$

3. **Probability of an event**

For any set A,

$$P((X, Y) \in A) = \sum_{(x,y) \in A} p_{X,Y}(x, y)$$

---

3. Joint PMF Table Representation

For discrete variables, the joint PMF is commonly represented in a **table format**.

Example structure:

Y\X	x1	x2	x3
y1	p(x1,y1)	p(x2,y1)	p(x3,y1)
y2	p(x1,y2)	p(x2,y2)	p(x3,y2)

Y\X	x1	x2	x3
y3	p(x1,y3)	p(x2,y3)	p(x3,y3)

Each table entry represents

$$P(X = x_i, Y = y_j)$$

The sum of **all entries in the table must equal 1**.

---

4. Example: Rolling Two Dice

Experiment:

Roll **two fair dice**.

Define random variables:

$$X = \text{value on the first die}$$

$$T = \text{total (sum) of both dice}$$

Possible values:

$$X \in \{1, 2, 3, 4, 5, 6\}$$

$$T \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

Total sample outcomes:

$$6 \times 6 = 36$$

Each outcome has probability

$$\frac{1}{36}$$

Constructing the Joint PMF

The joint PMF

$$p_{X,T}(x, t) = P(X = x, T = t)$$

depends on the pair (x, t).

For a fixed x, the total t depends on the second die.

Example calculations:

Case 1

$$P(X = 1, T = 2)$$

This occurs when

$$(1, 1)$$

Probability

$$\frac{1}{36}$$

Case 2

$$P(X = 1, T = 7)$$

This requires second die = 6

Outcome

$$(1, 6)$$

Probability

$$\frac{1}{36}$$

Case 3

$$P(X = 3, T = 7)$$

Second die must be

$$4$$

Outcome

$$(3, 4)$$

Probability

$$\frac{1}{36}$$

Joint Probability Table

For each pair (x, t), compute the probability from valid outcomes.

If a pair (x, t) **cannot occur**, then

$$p_{X,T}(x, t) = 0$$

Example:

$$P(X = 6, T = 3) = 0$$

because the smallest total with $X = 6$ is 7.

---

5. Continuous Case – Joint PDF

If X and Y are **continuous random variables**, we use the **Joint Probability Density Function (Joint PDF)**.

Definition

$$f_{X,Y}(x, y)$$

represents the joint density of X and Y.

The probability that (X, Y) lies inside a region R is

$$P((X, Y) \in R) = \int \int_R f_{X,Y}(x, y)\, dx\, dy$$

Properties of Joint PDF

1. Non-negativity

$$f_{X,Y}(x, y) \geq 0$$

2. Total probability

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx\, dy = 1$$

---

6. Geometric Interpretation

In continuous distributions, probability corresponds to **area or volume under a surface**.

Interpretation:

- The function $f_{X,Y}(x, y)$ forms a **3D surface**
- Probability over a region is the **volume under the surface**

Example probability:

$$P(a < X < b, c < Y < d)$$
$$= \int_a^b \int_c^d f_{X,Y}(x, y)\, dy\, dx$$

---

7. Worked Example (Joint PDF Normalization)

Suppose a joint PDF is defined as

$$f_{X,Y}(x, y) = c$$

inside a region R.

To determine c, use the normalization condition.

Step 1: Apply total probability condition

$$\int \int_R f_{X,Y}(x, y)\, dx\, dy = 1$$

Substitute the function:

$$\int \int_R c\, dx\, dy = 1$$

Step 2: Factor out constant

$$c \int \int_R dx\, dy = 1$$

Step 3: Compute area of region

If region bounds are

$$a < x < b$$

$$c < y < d$$

Area

$$(b-a)(d-c)$$

Step 4: Solve for constant

$$c(b-a)(d-c) = 1$$

$$c = \frac{1}{(b-a)(d-c)}$$

---

8. Joint Cumulative Distribution Function (Joint CDF)

The **Joint CDF** describes cumulative probability.

Definition

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$$

This represents the probability that

- X is less than or equal to x
- AND
- Y is less than or equal to y

---

9. Joint CDF – Continuous Case

For continuous variables,

$$F_{X,Y}(x,y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{X,Y}(u,v)\, dv\, du$$

Relation Between CDF and PDF

The joint PDF can be obtained by differentiation:

$$f_{X,Y}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x,y)$$

Thus, the PDF is the **second partial derivative of the CDF**.

10. Joint CDF – Discrete Case

For discrete random variables:

$$F_{X,Y}(x,y) = P(X \le x, Y \le y)$$

$$F_{X,Y}(x,y) = \sum_{i \le x} \sum_{j \le y} p_{X,Y}(i,j)$$

where

$$p_{X,Y}(i,j)$$

is the joint PMF.

---

11. Properties of Joint CDF

1. Range

$$0 \le F_{X,Y}(x,y) \le 1$$

2. Limits

$$\lim_{x \to -\infty} F(x,y) = 0$$

$$\lim_{y \to -\infty} F(x,y) = 0$$

3. Upper limit

$$\lim_{x,y \to \infty} F(x,y) = 1$$

4. Monotonicity

The CDF is **non-decreasing** in both variables.

Rectangle Probability Formula

Probability for region

$$a < X \le b, \quad c < Y \le d$$

is

$$P(a < X \le b, c < Y \le d)$$

$$= F(b,d) - F(a,d) - F(b,c) + F(a,c)$$

---

12. Key Formulas for Exam Revision

Joint PMF

$$p_{X,Y}(x,y) = P(X = x, Y = y)$$

$$\sum_x \sum_y p_{X,Y}(x,y) = 1$$

Joint PDF

$$P((X,Y) \in R) = \int \int_R f_{X,Y}(x,y)\, dx\, dy$$

$$\iint f_{X,Y}(x,y)\, dx\, dy = 1$$

Joint CDF

$$F_{X,Y}(x,y) = P(X \le x, Y \le y)$$

Continuous relation:

$$F(x,y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f(u,v)\, dv\, du$$

PDF from CDF:

$$f(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y)$$

If you want, I can also make a **2-page ultra-compressed "night-before-exam cheat sheet" for this lecture (with solving tricks your professor typically tests)** which will help you **revise the entire topic in ~5 minutes.**

generate a lecture scribe for this. lecture scribe is a exam ready material for before exam preparation. include important explanation, derivations and calculations given in the pdf. includes all the concepts covered in the pdf. don't give unnecessary repetitions and motivation. stick to the pdf only and do not include concepts outside the pdf and not in context with it. give detailed lecture scribe following all the rules

Below is a **structured lecture scribe strictly based on the provided slides** .
It includes **all concepts present in the lecture (Joint Random Variables, Joint PMF, Joint PDF, Joint CDF, properties, geometric interpretation, and worked examples)** with **derivations and calculations**, written as **exam-ready revision material**.
No external concepts or unnecessary motivation are added.

CSE400 – Fundamentals of Probability in Computing

Lecture Scribe (L15–L18): Joint Random Variables & Joint Distributions

---

1. Joint Random Variables

In many probabilistic systems we need to analyze **two random variables simultaneously**.

Let

$$X = \text{Random Variable } 1$$

$$Y = \text{Random Variable } 2$$

The ordered pair

$$(X, Y)$$

is called a **joint random variable**.

The objective is to determine probabilities involving **combinations of values** of both variables.

Examples mentioned in the lecture include:

- $X =$ vehicle speed, $Y =$ traffic density

- $X =$ SNR, $Y =$ data throughput
- $X =$ rainfall, $Y =$ temperature
- $X =$ wind speed, $Y =$ delivery time
- Dice experiments

Such situations require **joint probability distributions**.

---

2. Discrete Joint Distribution (Joint PMF)

If both X and Y are **discrete**, the joint behavior is represented using the **Joint Probability Mass Function**.

Definition

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

This gives the probability that

- $X = x$
- and $Y = y$

occur simultaneously.

Properties of Joint PMF

1. Non-negativity

$$p_{X,Y}(x, y) \geq 0$$

---

2. Total probability

The sum of all probabilities equals 1.

$$\sum_x \sum_y p_{X,Y}(x, y) = 1$$

---

3. Probability of a set

For a region A,

$$P((X, Y) \in A)$$

$$= \sum_{(x,y)\in A} p_{X,Y}(x,y)$$

---

3. Joint PMF Table Representation

For discrete variables, joint probabilities are often organized in a **table**.

Example structure:

Y \ X	x_1	x_2	x_3
y_1	$p(x_1,y_1)$	$p(x_2,y_1)$	$p(x_3,y_1)$
y_2	$p(x_1,y_2)$	$p(x_2,y_2)$	$p(x_3,y_2)$
y_3	$p(x_1,y_3)$	$p(x_2,y_3)$	$p(x_3,y_3)$

Each entry represents

$$P(X = x_i, Y = y_j)$$

Important property:

$$\text{Sum of all table entries} = 1$$

---

4. Example – Rolling Two Dice

Experiment:

Roll two fair dice.

Define

$$X = \text{value of the first die}$$

$$Y = \text{value of the second die}$$

Possible values:

$$X, Y \in \{1, 2, 3, 4, 5, 6\}$$

Sample Space

Total outcomes:

$$6 \times 6 = 36$$

Each outcome has probability

$$\frac{1}{36}$$

Joint PMF

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

Since the dice are fair:

$$p_{X,Y}(x, y) = \frac{1}{36}$$

for

$$x = 1..6, \quad y = 1..6$$

otherwise

$$p_{X,Y}(x, y) = 0$$

Example Calculation 1

Find

$$P(X + Y = 7)$$

Possible outcomes:

$$(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$$

Number of favorable outcomes

$$6$$

Total outcomes

$$36$$

$$P(X + Y = 7) = \frac{6}{36}$$

$$P(X+Y=7)=\frac{1}{6}$$

Example Calculation 2

Find

$$P(X>Y)$$

Possible outcomes:

(2,1)

(3,1),(3,2)

(4,1),(4,2),(4,3)

(5,1),(5,2),(5,3),(5,4)

(6,1),(6,2),(6,3),(6,4),(6,5)

Total outcomes

$$15$$

Probability

$$P(X>Y)=\frac{15}{36}$$
$$P(X>Y)=\frac{5}{12}$$

---

5. Continuous Joint Distribution (Joint PDF)

If X and Y are **continuous random variables**, their joint distribution is represented by a **Joint Probability Density Function (PDF)**.

Definition

$$f_{X,Y}(x,y)$$

represents the joint density.

The probability that (X,Y) lies in region R is

$$P((X,Y) \in R) = \int\int_R f_{X,Y}(x,y)\,dx\,dy$$

Properties

Non-negativity

$$f_{X,Y}(x,y) \geq 0$$

Total probability

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x,y)dx\,dy = 1$$

---

6. Geometric Interpretation

The joint PDF forms a **3-dimensional surface**.

Interpretation:

- x and y represent coordinates on a plane
- $f(x,y)$ represents the height of the surface
- Probability equals the **volume under the surface over a region**

Example probability:

$$P(a < X < b,\ c < Y < d)$$
$$= \int_a^b \int_c^d f_{X,Y}(x,y)\,dy\,dx$$

---

7. Worked Example (Normalization)

Suppose

$$f_{X,Y}(x,y) = c$$

inside a region R.

We must determine c.

Step 1 — Apply normalization condition

$$\int\int_R f_{X,Y}(x,y)\,dx\,dy = 1$$

Substitute

$$\int\int_R c\,dx\,dy = 1$$

Step 2 — Factor constant

$$c\int\int_R dx\,dy = 1$$

Step 3 — Compute region area

If

$$a < x < b$$

$$c < y < d$$

Area

$$(b-a)(d-c)$$

Step 4 — Solve for constant

$$c(b-a)(d-c) = 1$$

$$c = \frac{1}{(b-a)(d-c)}$$

---

8. Joint Cumulative Distribution Function (Joint CDF)

The joint CDF describes cumulative probability.

Definition

$$F_{X,Y}(x,y) = P(X \le x,\ Y \le y)$$

It represents the probability that

- X does not exceed x
- and Y does not exceed y

simultaneously.

---

9. Joint CDF – Continuous Case

For continuous random variables:

$$F_{X,Y}(x,y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{X,Y}(u,v)\, dv\, du$$

Relation between CDF and PDF

The joint PDF can be obtained by differentiating the CDF.

$$f_{X,Y}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x,y)$$

Thus the PDF is the **second partial derivative of the CDF**.

---

10. Joint CDF – Discrete Case

For discrete variables:

$$\begin{aligned} F_{X,Y}(x,y) &= P(X \le x, Y \le y) \\ &= \sum_{i \le x} \sum_{j \le y} p_{X,Y}(i,j) \end{aligned}$$

where

$$p_{X,Y}(i,j)$$

is the joint PMF.

---

11. Properties of the Joint CDF

Range

$$0 \le F_{X,Y}(x,y) \le 1$$

Lower limits

$$\lim_{x \to -\infty} F(x, y) = 0$$

$$\lim_{y \to -\infty} F(x, y) = 0$$

Upper limit

$$\lim_{x, y \to \infty} F(x, y) = 1$$

Monotonicity

The joint CDF is **non-decreasing** in both variables.

---

12. Probability Using Joint CDF

Probability over a rectangular region:

$$P(a < X \leq b,\ c < Y \leq d)$$

$$= F(b, d) - F(a, d) - F(b, c) + F(a, c)$$

This formula is derived by subtracting cumulative probabilities outside the rectangle.

---

13. Key Formulas for Exam Revision

Joint PMF

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

$$\sum_x \sum_y p_{X,Y}(x, y) = 1$$

Joint PDF

$$P((X, Y) \in R) = \int \int_R f_{X,Y}(x, y)\, dx\, dy$$

$$\iint f_{X,Y}(x,y)\,dx\,dy = 1$$

Joint CDF

$$F_{X,Y}(x,y) = P(X \le x, Y \le y)$$

Continuous case:

$$F(x,y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f(u,v)\,dv\,du$$

PDF from CDF:

$$f(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y)$$

If you want, I can also convert this into a **1–2 page "CSE400 exam cheat sheet" (only formulas, derivations, and solving patterns your professor typically tests)** so you can **revise this entire lecture in ~3–4 minutes before the exam.**